Linear Trigonometric Equations

Exam code: 9MA0

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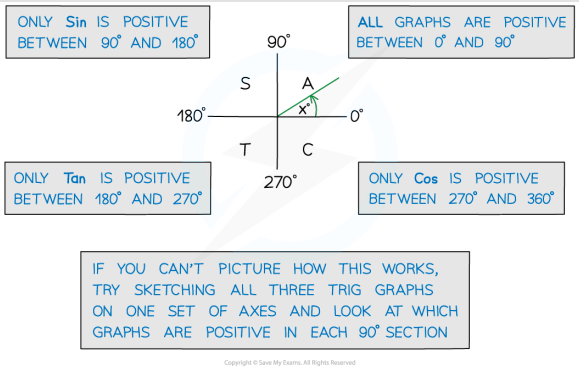
Linear trigonometric equations

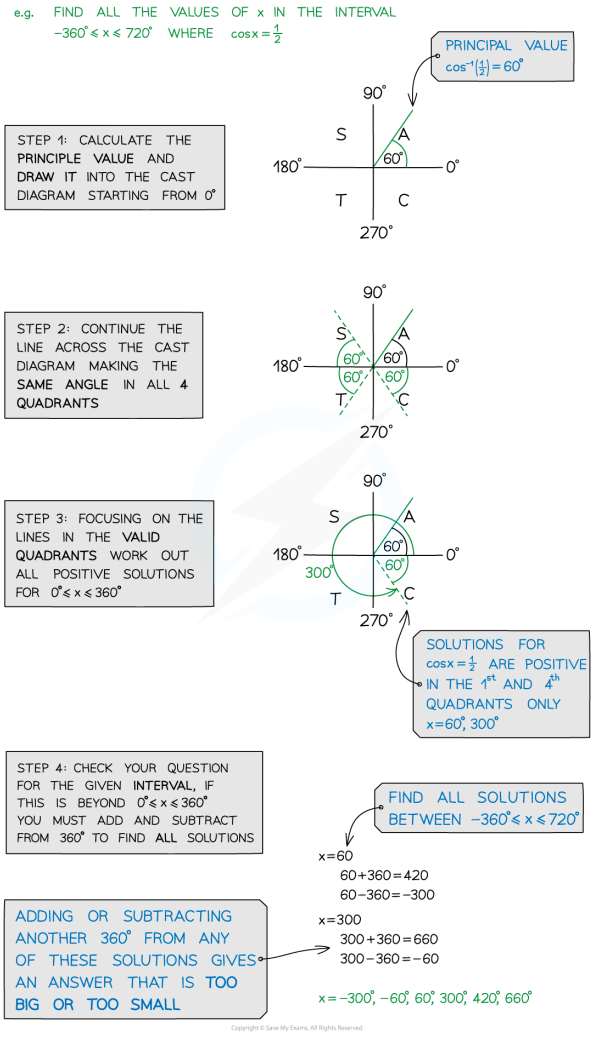
How do I solve linear trigonometric equations?

- You should have already seen two ways to solve linear trig equations
  - By **sketching the graph** (see Graphs of Trigonometric Functions) you can read off all the solutions in a given range (or interval)
  - By using **trigonometric identities** you can simplify harder equations
- Another way to find solutions is by using the CAST diagram which shows where each function has positive solutions
- You may be asked to use **degrees or radians** to solve trigonometric equations
  - Make sure your calculator is in the **correct mode**
  - Remember common angles
    - 90° is ½π radians
    - 180° is π radians
    - 270° is 3π/2 radians
    - 360° is 2π radians

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

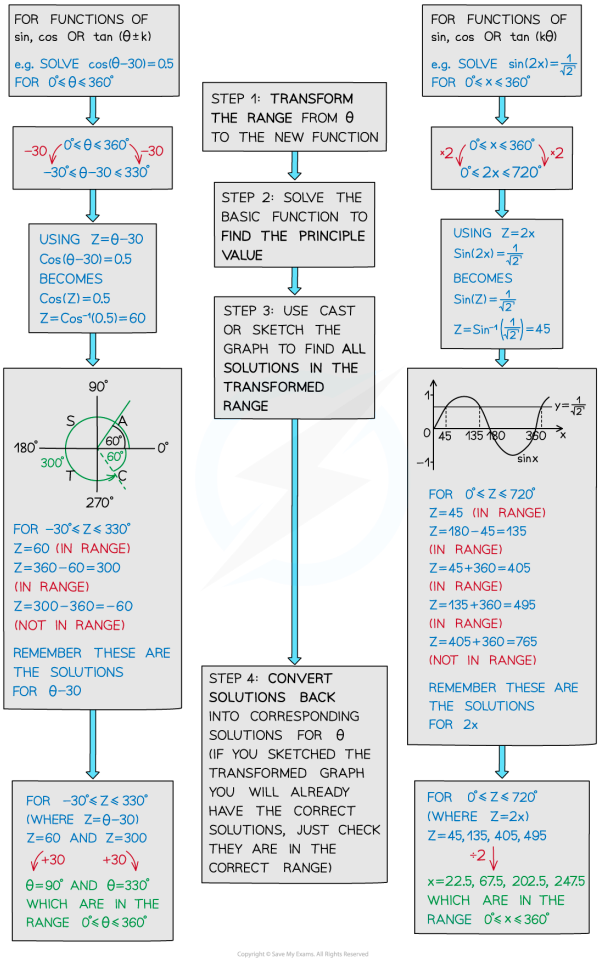


How do I use the CAST diagram?



How do I solve more complicated trigonometric equations?

- Some trig equations could involve a **function** of x or θ (see Transformations of Trigonometric Functions)
- Functions could be in two forms, either $y =$ **$\sin(\theta \pm k)$** or $y =$ **$\sin k\theta$**



Examiner Tips and Tricks

- Your calculator will only give you the principal value and you need to find all other solutions for the given interval.
- Also, remember the CAST diagram only gives you some solutions, so again you may need to find more depending on the given interval.
- It is entirely up to you how you solve a trig equation, but some ways are more helpful than others depending on the type of equation you are trying to solve

Worked Example

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